

**Saanidhya Shahi**

**09619051623**

**AIML B2**

**Fuzzy Logic**

## **Exp. 2 : Plotting of basic and membership functions.**

### **1. Plot Basic Mathematical Functions**

**Code:-**

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
# Define the range equivalent to -10:0.1:10 x
```

```
x = np.arange(-10, 10.1, 0.1)
```

```
# Close existing plots (equivalent to close all) plt.close('all')
```

```
# Create a figure
```

```
plt.figure(figsize=(8, 10))
```

```
# Subplot 1:  $y = x^2$ 
```

```
plt.subplot(3, 1, 1)
```

```
plt.plot(x, x**2) plt.title('y
```

```
= x^2') plt.grid(True)
```

```
# Subplot 2:  $y = \sin(x)$ 
```

```
plt.subplot(3, 1, 2)
```

```
plt.plot(x, np.sin(x))
```

```
plt.title('y = sin(x)')
```

```
plt.grid(True)
```

```
# Subplot 3:  $y = e^x$ 
```

```
plt.subplot(3, 1, 3)
```

```
plt.plot(x, np.exp(x))
```

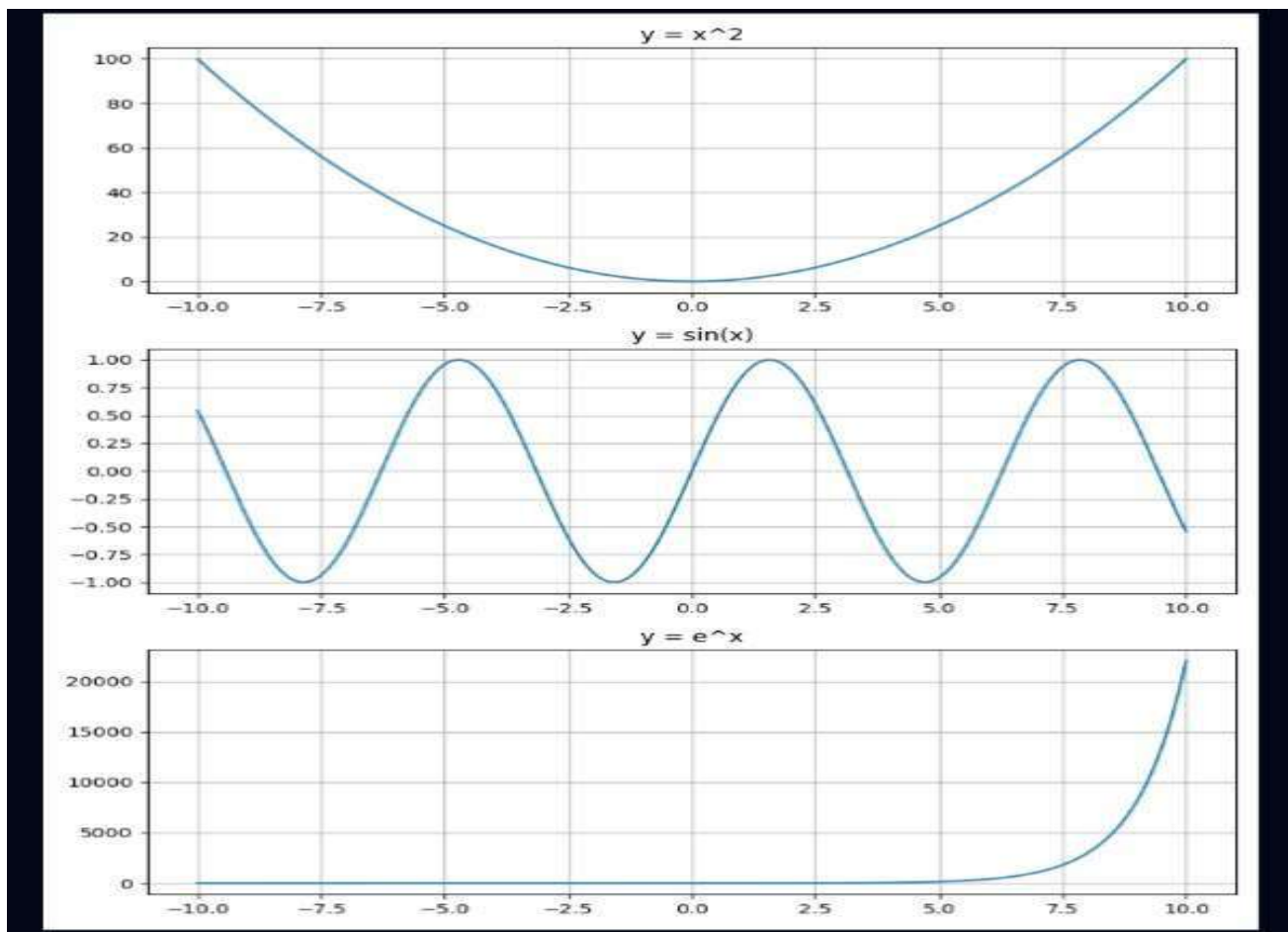
```
plt.title('y =  $e^x$ ')
```

```
plt.grid(True)
```

```
# Adjust layout and display plt.tight_layout()
```

```
plt.show()
```

**OUTPUT:-**



## 2. Plot Fuzzy Membership Functions

### a) Triangular Membership Function

#### Code:-

```
clc; clear;
```

```
close all;
```

```
x = 0:0.1:10; y =
```

```
trimf(x, [2 5 8]);
```

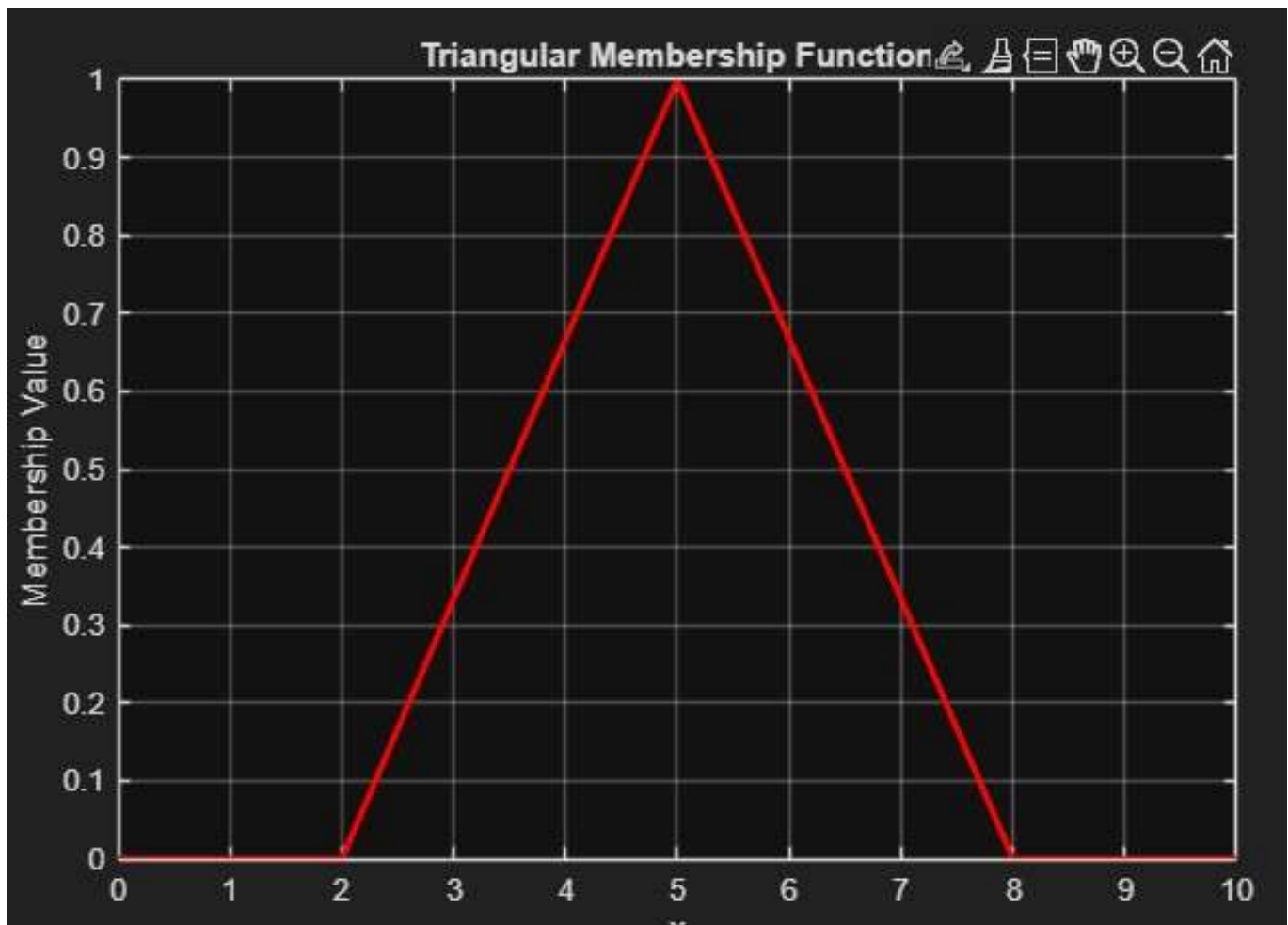
```
plot(x, y, 'r', 'LineWidth', 2);
```

```
title('Triangular Membership Function');
```

```
xlabel('x'); ylabel('Membership Value');
```

```
grid on;
```

#### OUTPUT:-

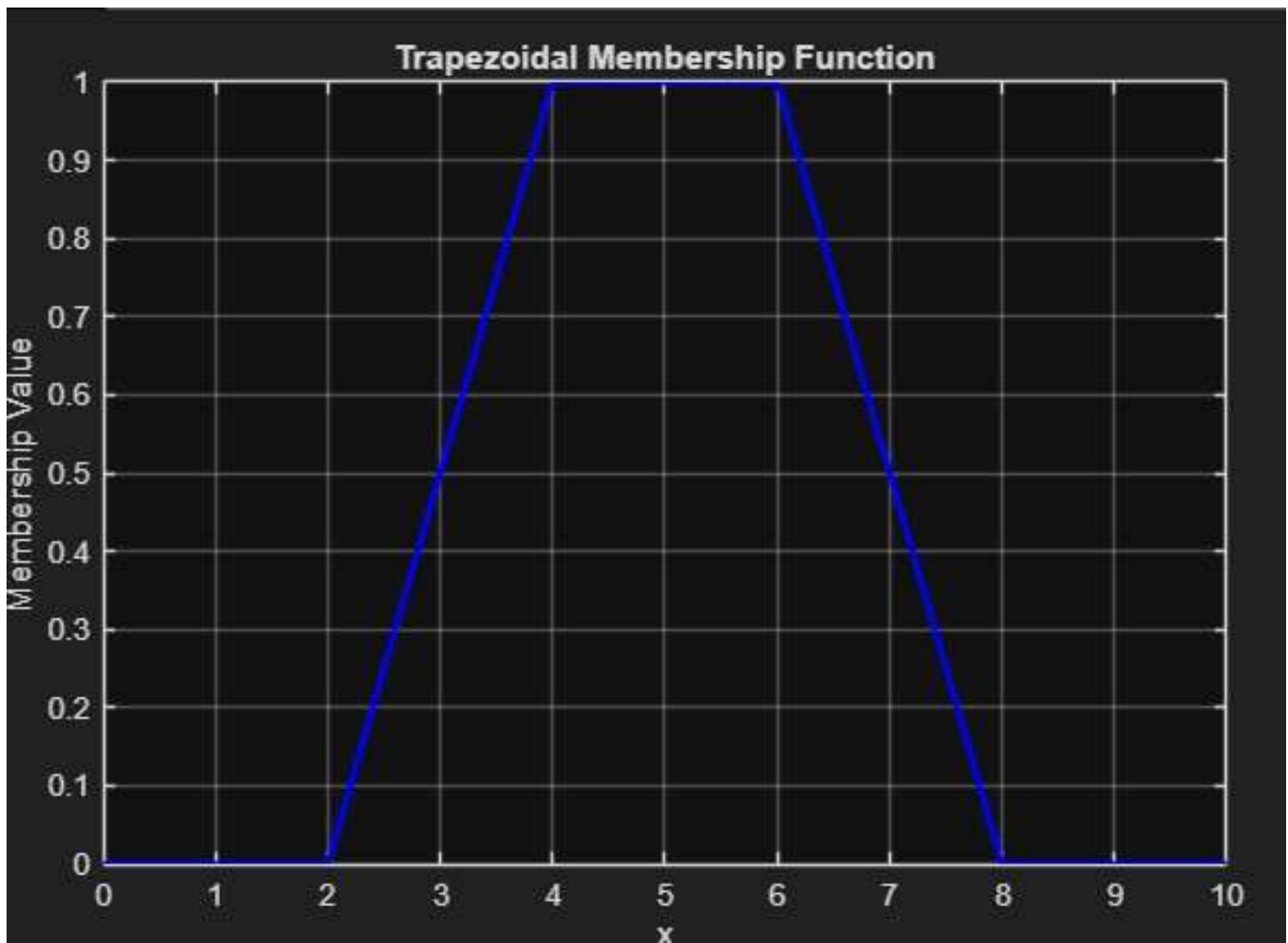


## b) Trapezoidal Membership Function

### Code:-

```
clc; clear;  
close all;  
  
x = 0:0.1:10;  
y = trapmf(x, [2 4 6 8]);  
  
plot(x, y, 'b', 'LineWidth', 2);  
title('Trapezoidal Membership Function');  
xlabel('x'); ylabel('Membership Value');  
grid on;
```

### OUTPUT:-



### c) Gaussian Membership Function

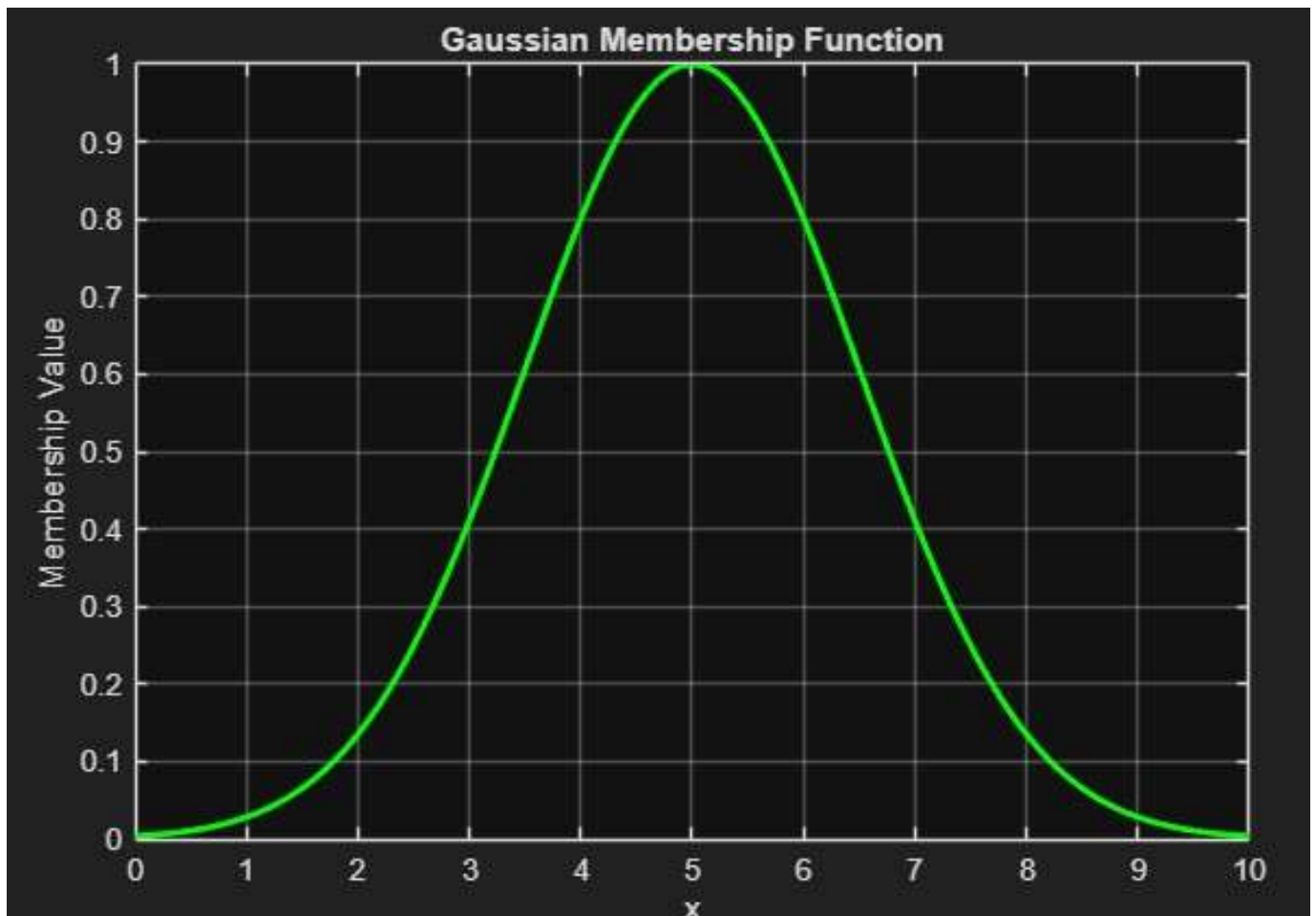
#### Code:-

```
clc; clear;  
close all;
```

```
x = 0:0.1:10; y = gaussmf(x, [1.5 5]); %  
[sigma mean]
```

```
plot(x, y, 'g', 'LineWidth', 2);  
title('Gaussian Membership Function');  
xlabel('x'); ylabel('Membership Value');  
grid on;
```

#### OUTPUT:-



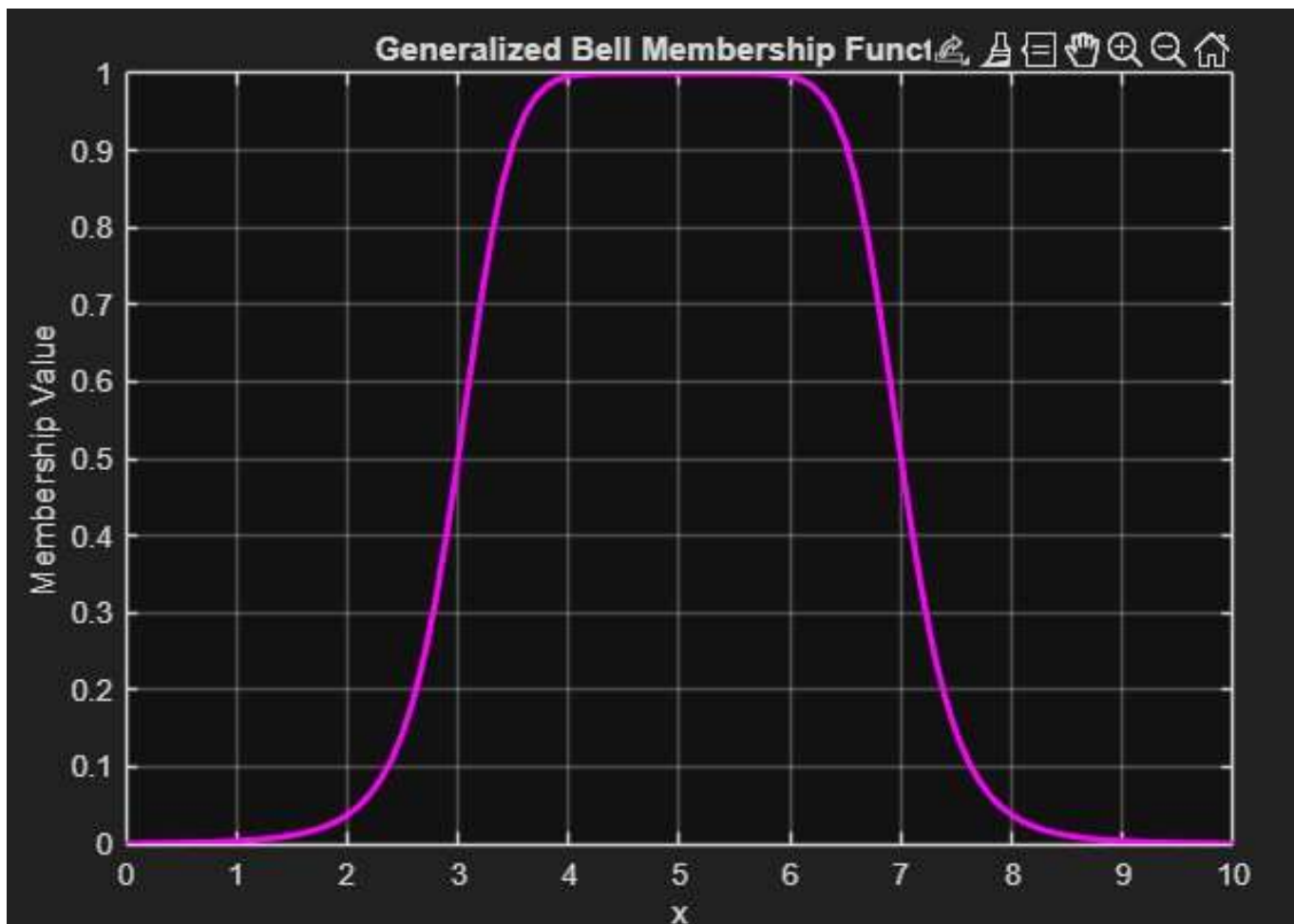
#### d) Generalized Bell Membership Function Code:-

```
clc; clear;  
close all;
```

```
x = 0:0.1:10; y = gbellmf(x, [2 4  
5]); % [a b c]
```

```
plot(x, y, 'm', 'LineWidth', 2);  
title('Generalized Bell Membership Function');  
xlabel('x'); ylabel('Membership Value'); grid  
on;
```

#### OUTPUT:-



## e) Sigmoid Membership Function

### Code:-

```
clc;
clear;
close all;

% Define range
x = -10:0.01:10;

% Parameters
a = 1; % slope
c = 0; % center

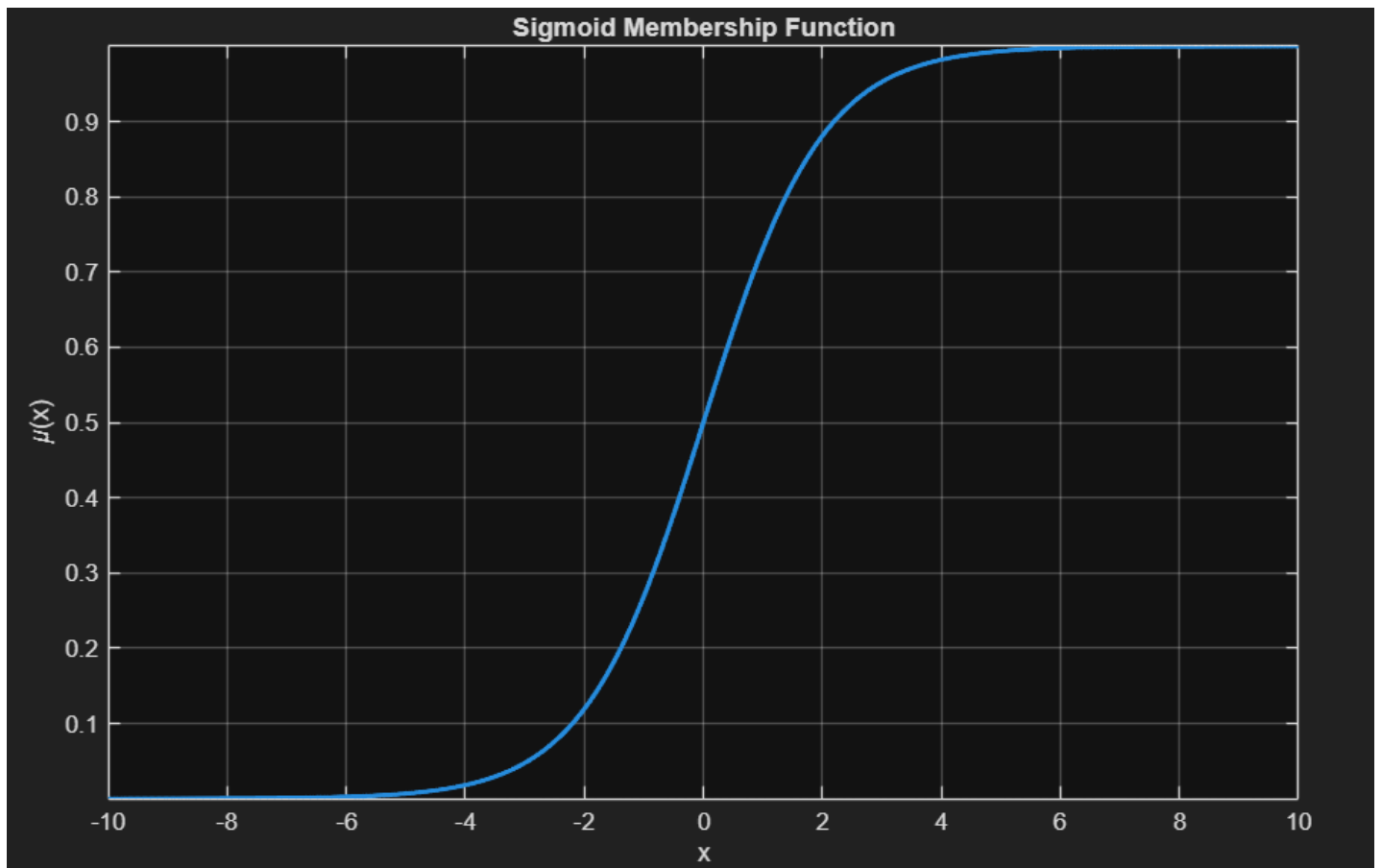
% Sigmoid Membership Function
y = 1 ./ (1 + exp(-a * (x - c)));

% Plot
plot(x, y, 'LineWidth', 2);
grid on;

% Labels
xlabel('x');
ylabel('\mu(x)');
title('Sigmoid Membership Function');

axis tight;
```

OUTPUT:-





### 3. Yaeger's Function

#### Code:-

```
clc;

clear;

close all;


% Define range

x = -5:0.01:5;


% Yaeger's Function

y = x .* exp(-x.^2);


% Plot

plot(x, y, 'LineWidth', 2);

grid on;


% Labels

xlabel('x');

ylabel('Y(x) = x * e^{-x^2}');

title('Yaeger Function Plot');


% Axis formatting

axis tight;
```

Output:-

